

Computer Vision Lab Exam — MCQ Cheat Sheet

Correct answers are highlighted directly under each question. No answer-key flipping required.

Practice Set 1

1. Which function reads an image from disk?

✓ A. cv2.imread

- B. cv2.readImage
- C. cv2.load
- D. cv2.open

2. cv2.imread normally returns color channels in which order?

A. RGB

✓ B. BGR

- C. HSV
- D. GRAY

3. Which conversion turns BGR into grayscale?

✓ A. cv2.COLOR_BGR2GRAY

- B. cv2.COLOR_RGB2GRAY
- C. cv2.COLOR_BGR2HSV
- D. cv2.GRAY2BGR

4. What is the usual 8-bit grayscale range?

A. 0-1

B. 0-100

✓ C. 0-255

D. 1-256

5. Which function changes an image size?

✓ A. cv2.resize

- B. cv2.scale
- C. cv2.changeSize
- D. cv2.reshapeImage

6. Which function creates a rotation matrix around a point?

✓ A. cv2.getRotationMatrix2D

- B. cv2.rotateMatrix
- C. cv2.rotationMatrix
- D. cv2.makeRotation

7. Which function applies a 2D affine matrix to an image?

✓ A. cv2.warpAffine

- B. cv2.warpPerspective
- C. cv2.applyAffineOnly
- D. cv2.affineWarp

8. Why convert BGR to RGB before normal color display with Matplotlib?

✓ A. Matplotlib normally expects RGB order

- B. It increases resolution
- C. It removes noise
- D. It makes the image grayscale

9. Which statement about NumPy image slicing is correct?

✓ A. It can crop/select rows and columns

- B. It detects circles
- C. It computes homography
- D. It creates SIFT descriptors

10. For img[100:400,100:500], the first range refers to:

A. Columns

✓ B. Rows

- C. Channels
- D. Color values Histogram and Enhancement CV Exam MCQ Cheat Sheet — Page 1

11. Which function computes a grayscale histogram in OpenCV?

✓ A. cv2.calcHist

- B. cv2.histogram
- C. cv2.getHistogram
- D. cv2.makeHist

12. Histogram equalization mainly improves:

- ✓ A. Contrast
- B. Image dimensions
- C. FPS
- D. Number of channels

13. Which function performs grayscale histogram equalization?

- ✓ A. cv2.equalizeHist
- B. cv2.histEqual
- C. cv2.equalize
- D. cv2.histNormalize

14. A histogram concentrated on the left usually indicates an image that is:

- A. Bright
- ✓ B. Dark
- C. Highly saturated
- D. Rotated

15. A histogram concentrated on the right usually indicates an image that is:

- A. Dark
- ✓ B. Bright
- C. Blurred
- D. Binary only

16. Which transform is useful for enhancing dark details?

- ✓ A. Log transform
- B. Translation
- C. Rotation
- D. Cropping

17. For $\gamma < 1$, the image is generally:

- ✓ A. Brightened
- B. Darkened
- C. Rotated
- D. Binarized

18. For $\gamma > 1$, the image is generally:

- A. Brightened
- ✓ B. Darkened
- C. Segmented by color
- D. Warped

19. Why use float values for a log transform?

- ✓ A. To safely perform floating-point arithmetic
- B. To change BGR to RGB
- C. To detect contours
- D. To increase FPS

20. Which enhancement changes intensity mapping without changing image geometry?

- ✓ A. Gamma correction
- B. Rotation
- C. Translation
- D. Perspective warp Thresholding and ROI

21. Which function performs binary thresholding?

- ✓ A. cv2.threshold
- B. cv2.Canny
- C. cv2.findContours
- D. cv2.HoughLinesP

22. Which threshold method chooses a threshold automatically from the histogram?

- ✓ A. Otsu
- B. SIFT
- C. Hough
- D. Watershed

23. When using Otsu, the threshold argument is commonly set to:

- ✓ A. 0
- B. 127
- C. 255
- D. -1

24. Which method handles spatially varying illumination better?

✓ A. Adaptive thresholding

- B. Fixed global threshold
- C. Affine transform
- D. SIFT

25. adaptiveThreshold blockSize should normally be:

A. Even

✓ B. Odd

- C. Negative
- D. A floating point number

Practice Set 2

26. In THRESH_BINARY, a pixel above the threshold is commonly assigned:

A. 0

B. 1

C. 127

✓ D. 255

27. Which function creates a polygon mask?

✓ A. cv2.fillPoly

- B. cv2.makeMask
- C. cv2.roiPolygon
- D. cv2.drawROI

28. Which value commonly represents the selected region in an 8-bit mask?

A. 0

B. 1

C. 128

✓ D. 255

29. How do you apply a grayscale ROI mask in the cheat-sheet code?

✓ A. cv2.bitwise_and(gray,gray,mask=mask)

- B. cv2.equalizeHist(gray,mask)
- C. cv2.Canny(gray,mask)
- D. cv2.resize(gray,mask)

30. Creating a mask without applying it means you have:

✓ A. Only defined the ROI mask

- B. Already extracted the ROI
- C. Detected edges
- D. Found contours Filtering, Edges and Contours

31. Which function detects edges with the Canny algorithm?

✓ A. cv2.Canny

- B. cv2.EdgeDetect
- C. cv2.findEdges
- D. cv2.HoughEdges

32. Why blur before Canny?

✓ A. To reduce noise

- B. To add color
- C. To create keypoints
- D. To resize the image

33. Which filter is especially useful for salt-and-pepper noise?

✓ A. Median blur

- B. Hough blur
- C. SIFT blur
- D. Otsu blur

34. Which function finds object boundaries after segmentation?

✓ A. cv2.findContours

- B. cv2.findEdgesOnly
- C. cv2.detectBoundary
- D. cv2.boundaries

35. What does cv2.contourArea return?

✓ A. Area of a contour

- B. Perimeter only
- C. Circle radius
- D. Line angle

36. What does cv2.boundingRect return?

✓ A. x,y,width,height

- B. x1,y1,x2,y2 only
- C. center,radius
- D. angle,scale

37. Which contour retrieval mode keeps only external contours?

✓ A. RETR_EXTERNAL

- B. RETR_INTERNAL
- C. RETR_OUTER_ONLY
- D. RETR_SIMPLE

38. A simple way to count segmented objects is:

✓ A. Count contours after filtering by area

- B. Count histogram bins
- C. Count Canny thresholds
- D. Count image channels

39. Why filter contours by area?

✓ A. To reject tiny unwanted blobs

- B. To rotate objects
- C. To calculate RGB values
- D. To find homography

40. Which operation is mainly intensity/edge based rather than geometric?

✓ A. Canny

- B. Rotation
- C. Translation
- D. Cropping Segmentation, Morphology and Watershed

41. In K-means image segmentation, K represents:

✓ A. Number of clusters

- B. Kernel size
- C. Number of channels
- D. Number of images

42. Why reshape RGB pixels to (-1,3) for K-means?

✓ A. Each pixel becomes one 3-value sample

- B. It rotates the image
- C. It creates contours
- D. It changes RGB to HSV

43. Why use float32 pixels for cv2.kmeans?

✓ A. It expects numerical floating-point samples

- B. It detects edges better
- C. It creates a mask
- D. It saves memory only

44. Opening consists of:

✓ A. Erosion then dilation

- B. Dilation then erosion
- C. Blur then threshold
- D. Threshold then Canny

45. Closing consists of:

✓ A. Dilation then erosion

- B. Erosion then dilation
- C. Canny then Hough
- D. SIFT then RANSAC

46. Opening is commonly used to:

✓ A. Remove small noise

- B. Fill small holes
- C. Detect lines
- D. Estimate homography

47. Closing is commonly used to:

- ✓ A. Fill small gaps/holes
- B. Remove every object
- C. Detect SIFT
- D. Calculate a histogram

48. Watershed is particularly useful for:

- ✓ A. Separating touching objects
- B. Changing brightness
- C. Finding circles only
- D. Rotating images

49. In cv2.watershed, pixels labeled -1 represent:

- ✓ A. Region boundaries
- B. The first object
- C. Background only
- D. Invalid input

50. Which operation commonly provides foreground markers for watershed?

- ✓ A. Distance transform plus thresholding
- B. Histogram equalization only
- C. Hough circles
- D. Gamma correction

Practice Set 3

51. Which method detects straight lines?

- ✓ A. Hough line transform
- B. Hough circle transform
- C. SIFT
- D. Watershed

52. Which function detects line segments with the probabilistic Hough transform?

- ✓ A. cv2.HoughLinesP
- B. cv2.HoughCircles
- C. cv2.findLines
- D. cv2.LineTransform

53. In HoughLinesP, minLineLength controls:

- ✓ A. Minimum accepted line length
- B. Minimum image width
- C. Minimum brightness
- D. Minimum circle radius

54. In HoughLinesP, maxLineGap controls:

- ✓ A. Maximum gap for joining line segments
- B. Maximum image height
- C. Maximum edge threshold
- D. Maximum FPS

55. np.pi/180 in a Hough call represents:

- ✓ A. One degree in radians
- B. One pixel
- C. One channel
- D. One FPS

56. Which method detects circles?

- ✓ A. cv2.HoughCircles
- B. cv2.HoughLinesP
- C. cv2.findContours only
- D. cv2.SIFT_create

57. In HoughCircles, param2 is mainly related to:

- ✓ A. Accumulator threshold for accepting circles
- B. Image width
- C. Number of channels
- D. Gamma

58. A typical lane pipeline is:

✓ A. Grayscale → blur → Canny → ROI → HoughLinesP

B. SIFT → gamma → circles

C. Otsu → MRI fusion → SIFT

D. Wavelet → histogram → rotation

59. Why use an ROI in lane detection?

✓ A. To focus on the road/lane region

B. To increase color depth

C. To calculate histogram

D. To detect SIFT

60. If HoughLinesP detects too many short lines, a useful adjustment is:

✓ A. Increase minLineLength

B. Set gamma to zero

C. Use addWeighted

D. Remove grayscale conversion Affine and Homography

61. How many point pairs are minimally needed for a 2D affine transform?

A. 2

✓ B. 3

C. 4

D. 8 CV Exam MCQ Cheat Sheet — Page 5

62. How many point pairs are minimally needed for a homography?

A. 2

B. 3

✓ C. 4

D. 8

63. Which function calculates an affine matrix from three point pairs?

✓ A. cv2.getAffineTransform

B. cv2.findHomography

C. cv2.warpAffine

D. cv2.getPerspective

64. Which function calculates a homography?

✓ A. cv2.findHomography

B. cv2.getAffineTransform

C. cv2.warpAffine

D. cv2.HoughLinesP

65. Which function applies an affine transform to an image?

✓ A. cv2.warpAffine

B. cv2.warpPerspective

C. cv2.applyAffine

D. cv2.transformAffineImage

66. Which function applies a perspective transform using H?

✓ A. cv2.warpPerspective

B. cv2.warpAffine

C. cv2.applyHomographyImage

D. cv2.perspectiveWarpOnly

67. Which transform is needed to model perspective distortion between planar views?

✓ A. Homography

B. Median filtering

C. Histogram equalization

D. Opening

68. Which statement is the best exam shortcut?

✓ A. 3 points → affine; 4 points → homography

B. 3 points → homography; 4 → affine

C. Both need 8 points

D. Neither needs point correspondences

69. RANSAC is commonly used with homography to:

✓ A. Reject outlier correspondences

B. Increase brightness

C. Detect circles

D. Resize images

70. A document photographed at an angle is a classic use case for:

✓ A. Homography and perspective warp

- B. Opening only
- C. Canny only
- D. K-means only SIFT, Matching and Panorama

71. What does SIFT provide for matching?

✓ A. Keypoints and descriptors

- B. Only a histogram
- C. Only contours
- D. Only line segments

72. Which call creates a SIFT detector?

✓ A. cv2.SIFT_create()

- B. cv2.createSIFT()
- C. cv2.SIFT()
- D. cv2.makeSIFT()

73. Why are descriptors needed?

✓ A. To compare local feature appearance

- B. To resize images
- C. To create binary masks
- D. To calculate FPS

74. Which matcher call supports the two-nearest-neighbor ratio test?

✓ A. BFMatcher().knnMatch(...,k=2)

- B. HoughLinesP(...,k=2)
- C. findContours(...,k=2)
- D. equalizeHist(...,k=2) CV Exam MCQ Cheat Sheet — Page 6

75. Why compare the best and second-best SIFT matches?

✓ A. To reject ambiguous matches

- B. To find two circles
- C. To calculate two histograms
- D. To double image size

Practice Set 4

76. A common ratio threshold in the cheat-sheet code is:

- A. 0.25
- B. 0.50

✓ C. 0.75

- D. 1.50

77. A smaller descriptor distance generally means:

✓ A. A more similar descriptor match

- B. A larger object
- C. A brighter pixel
- D. A longer line

78. After good SIFT matches, which model maps one planar view to another?

✓ A. Homography

- B. Gamma
- C. Histogram
- D. Watershed

79. A basic panorama pipeline is:

✓ A. SIFT → matching → ratio test → homography → warp

- B. Otsu → circles → gamma → K-means
- C. Opening → closing → video
- D. Histogram → FPS → threshold

80. Why does a panorama need overlap between images?

✓ A. Overlap supplies common features for matching

- B. Overlap increases grayscale range
- C. Overlap is required for gamma
- D. Overlap creates circles Medical Imaging, Fusion and Wavelets

81. Which function performs weighted CT/MRI fusion in the cheat sheet?

✓ A. cv2.addWeighted

- B. cv2.addFusion
- C. cv2.fuseImages
- D. cv2.weightedBlend

82. Which call gives 70% CT and 30% MRI?

✓ A. cv2.addWeighted(ct,0.7,mri,0.3,0)

- B. cv2.addWeighted(ct,0.3,mri,0.7,0)
- C. cv2.threshold(ct,mri,0.7)
- D. np.maximum(ct,mri,0.7)

83. Before weighted fusion, CT and MRI should have:

✓ A. Compatible dimensions and types

- B. Different dimensions
- C. Different FPS
- D. Only RGB channels

84. What does np.maximum(a,b) do pixelwise for same-sized grayscale images?

✓ A. Selects the larger intensity

- B. Averages intensities
- C. Subtracts images
- D. Rotates images

85. Which library is used for the wavelet example?

✓ A. PyWavelets (pywt)

- B. Pandas
- C. Flask
- D. TensorFlow only

86. In a 2D Haar decomposition, LL is:

✓ A. Low-frequency approximation

- B. Horizontal detail only
- C. Vertical detail only
- D. Diagonal detail only

87. Which components contain high-frequency detail?

✓ A. LH, HL, HH

- B. LL only
- C. RGB only
- D. XYZ only

88. A simple wavelet anomaly detector can flag:

✓ A. Large detail coefficients

- B. Only bright pixels
- C. Only image borders
- D. FPS changes

89. Which task is especially suited to X-ray contrast enhancement?

✓ A. Histogram equalization or log/gamma enhancement

- B. Hough circles only
- C. VideoWriter
- D. pointPolygonTest

90. Which thresholding approach can be useful on an enhanced X-ray with uneven illumination?

✓ A. Adaptive thresholding

- B. Only fixed 127 threshold
- C. HoughLinesP
- D. SIFT matching

91. Which class reads video frames?

✓ A. cv2.VideoCapture

- B. cv2.VideoReader
- C. cv2.FrameReader
- D. cv2.readVideo

92. What does cap.read() return?

✓ A. ret and frame

- B. frame and FPS
- C. image and mask
- D. x and y

93. When `ret` is `False`, the normal response is to:

✓ A. Stop because no frame was read

- B. Run SIFT
- C. Increase gamma
- D. Run `HoughCircles`

94. Which call releases a video capture?

✓ A. `cap.release()`

- B. `cap.closeVideo()`
- C. `cv2.releaseVideo()`
- D. `cap.stop()`

95. Which call closes OpenCV windows?

✓ A. `cv2.destroyAllWindows()`

- B. `cv2.closeWindows()`
- C. `cv2.releaseWindows()`
- D. `plt.destroyVideo()`

96. Which key is used to quit in the cheat-sheet video loop?

✓ A. `q`

- B. `x`
- C. `e`
- D. `z`

97. Simple frame-difference motion detection uses:

✓ A. `cv2.absdiff` between consecutive frames

- B. Histogram equalization only
- C. Hough circles
- D. SIFT only

98. Why threshold the frame difference?

✓ A. To create a binary motion mask

- B. To estimate homography
- C. To create descriptors
- D. To increase FPS

99. Which NumPy function gives the mean intensity?

✓ A. `np.mean`

- B. `np.avglImage`
- C. `np.meanValue`
- D. `np.averageOnly`

100. Which NumPy function gives standard deviation?

✓ A. `np.std`

- B. `np.standard`
- C. `np.dev`

D. `np.stdevImage` CV Exam MCQ Cheat Sheet — Page 8 Twisted Exam-Style Questions

Practice Set 5

101. You made a polygon mask but the outside is not black. What is most likely missing?

✓ A. Applying the mask with `bitwise_and`

- B. Calling `equalizeHist` twice
- C. Using SIFT
- D. Calling `VideoWriter`

102. Lighting changes from one side of an image to another. Which segmentation choice is safer?

✓ A. Adaptive threshold

- B. Fixed global threshold only
- C. `HoughCircles`
- D. Homography

103. Two coins touch after thresholding. Which method is designed to separate them?

✓ A. Watershed

- B. Gamma correction
- C. Histogram equalization
- D. HSV threshold

104. Many SIFT matches are wrong. Which combination is strongest?

✓ A. Ratio test + RANSAC homography

- B. Global threshold + closing
- C. Gamma + histogram
- D. Hough circles + opening

105. A road image contains buildings and wires. Why restrict Hough detection to a lane ROI?

✓ A. To reduce irrelevant line detections

- B. To increase bit depth
- C. To change BGR to RGB
- D. To create descriptors

106. CT and MRI images have different dimensions. What should happen before direct addWeighted fusion?

✓ A. Resize/alignment to compatible dimensions

- B. Run HoughCircles
- C. Convert to video
- D. Use pointPolygonTest

107. A perspective warp leaves black areas around the result. Why?

✓ A. The output canvas contains locations with no source pixels

- B. SIFT always creates black pixels
- C. Canny forces black borders
- D. RANSAC deletes pixels

108. Four corner correspondences are given to rectify a tilted image. Which pair is central?

✓ A. findHomography + warpPerspective

- B. findContours + equalizeHist
- C. Canny + HoughCircles
- D. K-means + opening

109. A binary image has isolated tiny foreground dots. Which is usually better?

✓ A. Opening

- B. Closing
- C. Homography
- D. SIFT

110. A binary object has tiny breaks that should be connected. Which is usually better?

✓ A. Closing

- B. Opening
- C. Canny
- D. Hough What Should I Use?

111. Question says: "enhance contrast." Best first choice?

✓ A. Histogram equalization

- B. HoughCircles
- C. Watershed
- D. VideoCapture

112. Question says: "image is very dark." Best candidates?

✓ A. Log/gamma/equalization

- B. HoughLinesP only
- C. pointPolygonTest
- D. SIFT only CV Exam MCQ Cheat Sheet — Page 9

113. Question says: "uneven illumination."

✓ A. Adaptive threshold

- B. Fixed threshold only
- C. Affine transform
- D. Hough circles

114. Question says: "automatic threshold."

✓ A. Otsu

- B. SIFT
- C. Hough
- D. Gamma

115. Question says: "detect edges."

✓ A. Canny

- B. K-means
- C. Homography
- D. addWeighted

116. Question says: “detect straight lane markings.”

✓ **A. HoughLinesP**

- B. HoughCircles
- C. SIFT
- D. Watershed

117. Question says: “detect circular objects.”

✓ **A. HoughCircles**

- B. HoughLinesP
- C. Canny only
- D. equalizeHist

118. Question says: “divide pixels into K groups.”

✓ **A. K-means**

- B. RANSAC
- C. SIFT
- D. Watershed only

119. Question says: “recognize the same planar object in another image.”

✓ **A. SIFT + matching + homography**

- B. Otsu only
- C. Opening only
- D. Gamma only

120. Question says: “stitch overlapping images.”

✓ **A. Feature matching + homography + warping**

- B. Thresholding only
- C. Closing only
- D. Mean intensity only Parameter and Code Traps

121. In cv2.Canny(gray,50,150), the two numbers are mainly:

✓ **A. Lower and upper edge thresholds**

- B. Width and height
- C. Min/max radius
- D. HSV lower/upper

122. In cv2.HoughLinesP(edges,1,np.pi/180,50,...), the 1 is the:

✓ **A. Distance resolution in pixels**

- B. Number of channels
- C. Image width
- D. FPS

123. In cv2.HoughLinesP, np.pi/180 is the:

✓ **A. Angle resolution in radians**

- B. Line width
- C. Circle radius
- D. Gamma

124. In cv2.HoughCircles, minDist controls:

✓ **A. Minimum distance between circle centers**

- B. Minimum line length
- C. Minimum image width
- D. Minimum threshold

125. In cv2.HoughCircles, minRadius controls:

✓ **A. Smallest allowed circle radius**

- B. Smallest line gap
- C. Smallest contour area
- D. Smallest image dimension CV Exam MCQ Cheat Sheet — Page 10

Practice Set 6

126. In cv2.findHomography(src,dst,cv2.RANSAC,5.0), 5.0 is commonly:

✓ **A. RANSAC reprojection threshold**

- B. Image width
- C. Number of matches
- D. Gamma

127. In `cv2.addWeighted(a,alpha,b,beta,gamma)`, the last gamma is:

✓ A. Scalar added to the weighted sum

- B. Gamma correction exponent
- C. HSV hue
- D. Homography parameter

128. In `cv2.resize(img,(500,400))`, the tuple is:

✓ A. New width, new height

- B. New height, new width
- C. Rows, columns
- D. x,y coordinates

129. In `np.zeros(gray.shape,dtype=np.uint8)`, the dtype is used because a mask is:

✓ A. An 8-bit image

- B. A float descriptor
- C. A video
- D. A homography

130. In `cv2.fillPoly(mask,[points],255)`, 255 means:

✓ A. Fill the polygon with white/selected pixels

- B. Use a threshold of 255
- C. Rotate by 255 degrees
- D. Use 255 points

131. Best pipeline for lane detection under noise?

✓ A. Gray → Gaussian blur → Canny → ROI → HoughLinesP

- B. SIFT → Otsu → fusion
- C. K-means → circles → gamma
- D. Watershed → video writer

132. Best pipeline for recognizing a template under perspective change?

✓ A. SIFT → ratio test → RANSAC homography → perspective transform

- B. Otsu → opening → closing
- C. Canny → HoughCircles only
- D. Histogram → gamma only

133. Best pipeline for separating touching bright objects in a dark image?

✓ A. Threshold → morphology → distance transform → watershed

- B. SIFT → homography
- C. HSV → HoughLinesP
- D. Gamma → VideoCapture

134. Best simple approach for a red object on a varied background?

✓ A. HSV threshold + mask

- B. Grayscale only
- C. Homography only
- D. Wavelet only

135. If an image has salt-and-pepper noise before edge detection, use:

✓ A. Median blur then Canny

- B. HoughCircles then SIFT
- C. Gamma then VideoWriter
- D. Homography then Otsu

136. If contours detect hundreds of tiny noise blobs, first simple fix?

✓ A. Filter by contour area and/or use opening

- B. Use more SIFT features
- C. Increase panorama width
- D. Use addWeighted

137. If SIFT has fewer than four good matches, homography estimation is:

✓ A. Not reliably possible with the basic minimum requirement

- B. Guaranteed accurate
- C. Automatically replaced by affine
- D. Unnecessary

138. If the exam asks for 3 matching points and “shear/rotation/translation,” choose:

✓ A. Affine transformation

- B. Homography
- C. Watershed
- D. Hough CV Exam MCQ Cheat Sheet — Page 11

139. If the exam asks for 4 matching corners and “perspective,” choose:

✓ A. Homography

- B. Median filter
- C. Opening
- D. Canny

140. If the exam asks to save processed frames to an MP4, use:

✓ A. cv2.VideoWriter

- B. cv2.VideoCapture only
- C. cv2.calcHist
- D. cv2.findHomography Rapid-Fire Viva-to-MCQ Theory

141. Which transformation can include translation, rotation, scaling and shear?

✓ A. Affine

- B. Histogram equalization
- C. Watershed
- D. HSV threshold

142. Which transformation can represent perspective effects?

✓ A. Projective/homography

- B. Median blur
- C. Opening
- D. Canny

143. Which method is robust to some incorrect correspondences?

✓ A. RANSAC

- B. Otsu
- C. K-means
- D. Gamma

144. Which method is invariant-ish to scale and rotation and useful for local feature matching?

✓ A. SIFT

- B. Otsu
- C. HoughCircles
- D. Opening

145. Which method is unsupervised clustering?

✓ A. K-means

- B. Canny
- C. RANSAC
- D. SIFT

146. Which operation is erosion followed by dilation?

✓ A. Opening

- B. Closing
- C. Watershed
- D. Homography

147. Which operation is dilation followed by erosion?

✓ A. Closing

- B. Opening
- C. Canny
- D. SIFT

148. Which segmentation method uses local neighborhoods for threshold selection?

✓ A. Adaptive thresholding

- B. Fixed threshold
- C. Homography
- D. SIFT

149. Which transform converts a point from one planar view using H?

✓ A. cv2.perspectiveTransform

- B. cv2.pointPolygonTest
- C. cv2.equalizeHist
- D. cv2.calcHist

150. Which function checks whether a point is inside a polygon?

✓ A. cv2.pointPolygonTest

- B. cv2.findContours
- C. cv2.fillPoly
- D. cv2.inRange CV Exam MCQ Cheat Sheet — Page 12

Last-Minute Memory Strip

3 points → Affine

4 points → Homography

Uneven lighting → Adaptive threshold

Automatic threshold → Otsu

Remove small noise → Opening

Fill small gaps/holes → Closing

Touching objects → Watershed

Edges → Canny

Lines → HoughLinesP

Circles → HoughCircles

Feature matching → SIFT + ratio test

Reject bad matches → RANSAC

Apply H → warpPerspective

Apply ROI → bitwise_and + mask